```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import
train_test_split
from sklearn.ensemble import
RandomForestClassifier
from sklearn.metrics import accuracy_score,
classification_report, confusion_matrix,
roc_curve, auc
from sklearn.preprocessing import
LabelEncoder, label_binarize

np.random.seed(42)

# ============================
# LOAD DATA
# ============================
df = pd.read_csv("student_wellbeing.csv")

print("\nClass Distribution:")
print(df["stress_level"].value_counts())

print("\nClass Percentage:")
print(df["stress_level"].value_counts(normaliz
e=True) * 100)

numeric_cols = [
    "sleep_hours",
    "study_hours",
    "screen_time",
    "attendance",
    "anxiety_level",
    "academic_pressure",
    "social_support",
    "physical_activity"
]

# ============================
# VISUALIZATION FUNCTIONS
# ============================
def show_stress(data, title):
```

```
def show_stress(data, title):
    counts =
data["stress_level"].value_counts()
    plt.figure()
    plt.bar(counts.index, counts.values)
    plt.title(title)
    plt.show()

def show_missing(data, title):
    missing = data.isnull().sum()
    plt.figure()
    plt.bar(missing.index, missing.values)
    plt.title(title)
    plt.xticks(rotation=45)
    plt.show()


def show_box(data, title):
    plt.figure()
    plt.boxplot([data[col].dropna() for col in
numeric_cols], labels=numeric_cols)
    plt.title(title)
    plt.xticks(rotation=45)
    plt.show()


# ============================
# BEFORE CLEANING
# ============================
print("==== BEFORE CLEANING ====")
print("Missing:\n", df.isnull().sum())
print("Duplicates:", df.duplicated().sum())

show_stress(df, "Stress Before Cleaning")
show_missing(df, "Missing Before Cleaning")
show_box(df, "Outliers Before Cleaning")

# ============================
# DATA CLEANING
# ============================
df = df.drop_duplicates()

for col in numeric_cols:
```

```
    df[col] = df[col].fillna(df[col].median())

df["stress_level"] =
df["stress_level"].fillna(df["stress_level"].m
ode()[0])

df["sleep_hours"] = df["sleep_hours"].clip(3,
10)
df["study_hours"] = df["study_hours"].clip(0,
10)
df["screen_time"] = df["screen_time"].clip(1,
12)
df["attendance"] = df["attendance"].clip(30,
100)
df["anxiety_level"] =
df["anxiety_level"].clip(1, 10)
df["academic_pressure"] =
df["academic_pressure"].clip(1, 10)
df["social_support"] =
df["social_support"].clip(1, 10)
df["physical_activity"] =
df["physical_activity"].clip(0, 10)

# ============================
# AFTER CLEANING
# ============================
print("\n==== AFTER CLEANING ====")
print("Missing:\n", df.isnull().sum())
print("Duplicates:", df.duplicated().sum())

show_stress(df, "Stress After Cleaning")
show_missing(df, "Missing After Cleaning")
show_box(df, "Outliers After Cleaning")

# ============================
# FEATURE ENGINEERING
# ============================
df["screen_study_ratio"] = df["screen_time"] /
(df["study_hours"] + 1)
df["pressure_index"] = df["anxiety_level"] +
df["academic_pressure"]
df["health_score"] = df["sleep_hours"] +
```

```
print("\nFeatures added: screen_study_ratio,
pressure_index, health_score,
support_attendance_score")

# ============================
# ENCODING
# ============================
X = df.drop("stress_level", axis=1)
y = df["stress_level"]

encoder = LabelEncoder()
y = encoder.fit_transform(y)

# ============================
# TRAIN MODEL
# ============================
X_train, X_test, y_train, y_test =
train_test_split(
    X, y, test_size=0.2, random_state=42,
stratify=y
)

model = RandomForestClassifier(
    n_estimators=700,
    max_depth=15,
    min_samples_split=3,
    min_samples_leaf=1,
    class_weight="balanced",
    random_state=42
)

model.fit(X_train, y_train)

pred = model.predict(X_test)
acc = accuracy_score(y_test, pred)

print("\nAccuracy:", round(acc, 2))

# ============================
# CLASSIFICATION REPORT
# ============================
print("\nClassification Report:")
print(classification_report(y_test, pred,
```

```
# ============================
# CONFUSION MATRIX
# ============================
cm = confusion_matrix(y_test, pred)
print("\nConfusion Matrix:")
print(cm)

# ============================
# ROC CURVE
# ============================
y_score = model.predict_proba(X_test)
y_test_bin = label_binarize(y_test,
classes=[0, 1, 2])

plt.figure(figsize=(8, 6))

for i in range(len(encoder.classes_)):
    fpr, tpr, _ = roc_curve(y_test_bin[:, i],
y_score[:, i])
    roc_auc = auc(fpr, tpr)
    plt.plot(fpr, tpr,
label=f"{encoder.classes_[i]} AUC =
{roc_auc:.2f}")

plt.plot([0, 1], [0, 1], linestyle="--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.show()


# ============================
# SAVE MODEL
# ============================
import pickle

with open("model.pkl", "wb") as f:
    pickle.dump(model, f)

with open("encoder.pkl", "wb") as f:
    pickle.dump(encoder, f)
```



May 10, 2026

Predicting Student Stress

Predicting Student Stress Levels with Machine Learning

Problem Statement

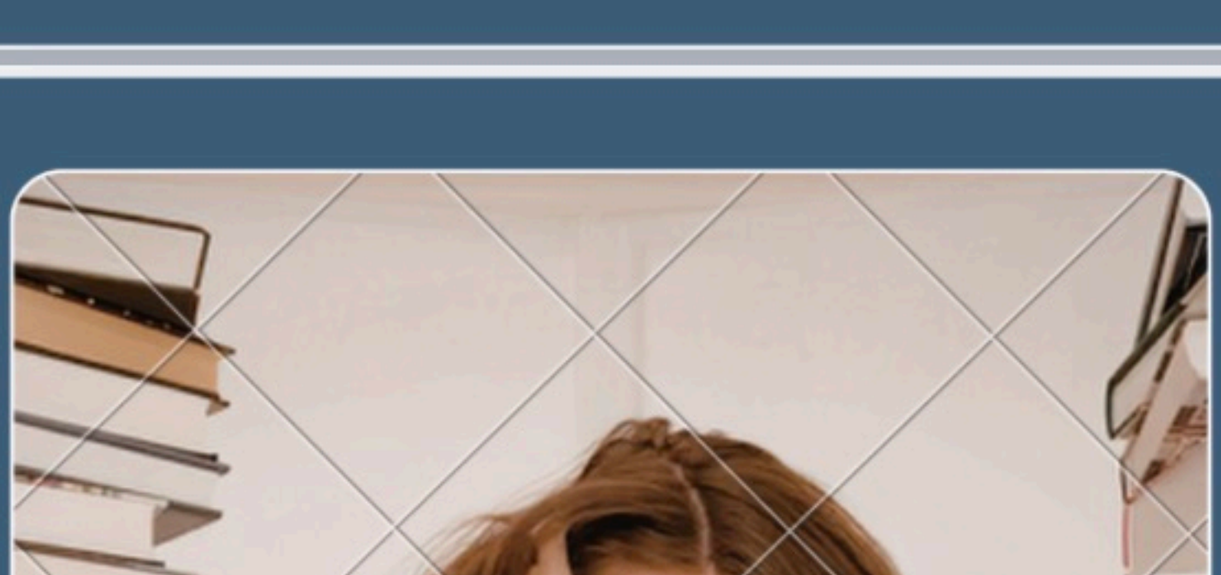
Students face significant stress influenced by various factors, impacting their academic performance and overall well-being.



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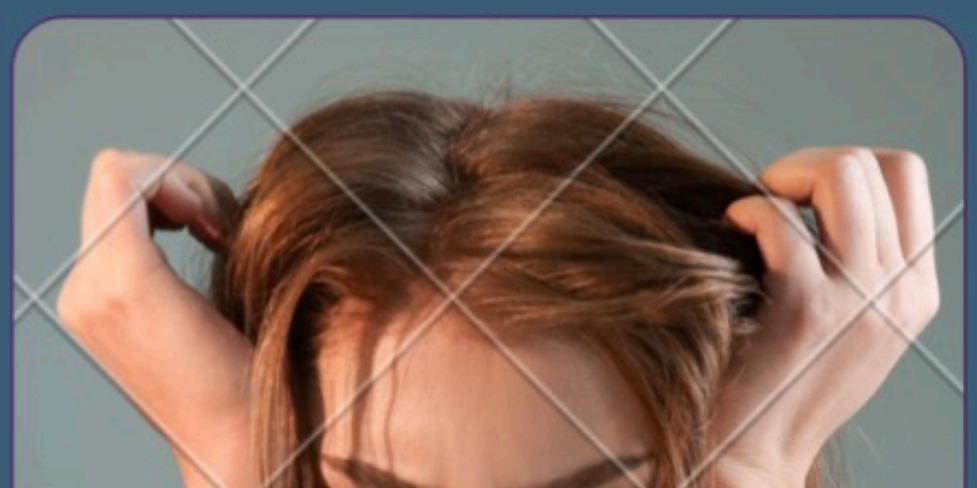
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